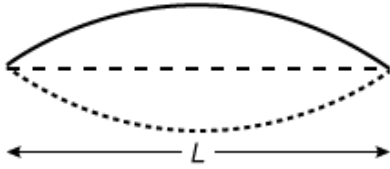


An elastic cord vibrates in the fundamental resonance between two fixed posts separated by a distance L as shown.



How will the fundamental frequency f change if the tension in the cord is increased?

- ☐ A. f will decrease because waves will travel at a lower speed along the cord. [14%]
- ☐ B. f will not change because the posts remain the same distance apart. [21%]
- ☐ C. f will not change because waves will travel at the same speed along the cord. [9%]
- ✓ ☒ D. f will increase because waves will travel at a higher speed along the cord. [54%]

Correct

55% answered correctly

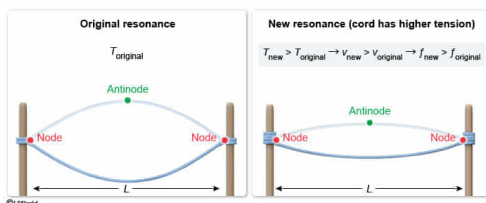
Explanation:

Increasing cord's tension increases frequency of fundamental resonance

$$f = \frac{v}{\lambda} \quad \lambda = 2L$$

v increases as T increases

- T Cord tension
- f Frequency
- L Length
- v Speed of waves
- λ Wavelength



Standing waves on a cord are generated due to interference between incident waves and reflected waves, which move in opposite directions along the cord. The resonating standing waves create nodes (points of zero amplitude) and antinodes (points of maximal amplitude). At the **fundamental resonance**, there is one antinode, and the **wavelength** λ of the standing wave is twice the length of the cord L :

$$\lambda = 2L$$

The frequency f of the wave is the ratio of the **wave speed** v to the wavelength:

$$f = \frac{v}{\lambda} = \frac{v}{2L}$$

Furthermore, the square of the speed of a wave on a cord is **directly proportional** to the **tension** force T , ie:

$$v \propto \sqrt{T}$$

In this question, when the tension in the cord is increased, the wave speed for the new wave is *greater* due to *increased* tension in the cord:

$$T_{\text{new}} > T_{\text{original}} \Rightarrow v_{\text{new}} > v_{\text{original}}$$

Moreover, the standing waves are in the fundamental resonance, with one antinode on the cord, which is fixed to posts that are the same distance apart. Therefore, the wavelength λ_{new} of the new wave is equal to that of the original wavelength $\lambda_{\text{original}}$:

$$\lambda_{\text{new}} = \lambda_{\text{original}} = 2L$$

Consequently,

$$\frac{v_{\text{new}}}{2L} > \frac{v_{\text{original}}}{2L}$$

$$f_{\text{new}} > f_{\text{original}}$$

At a constant wavelength, increasing speed will result in a greater frequency of the standing wave on the cord. Therefore, the second wave's frequency increases because the speed of the wave increases with tension.

(Choices A and C) Waves on the cord travel with greater speed because the cord has a greater tension, leading to an increase in frequency.

(Choice B) The length of the cord is fixed but an increase in tension also increases wave speed and frequency.

Educational objective:

The speed of waves on a cord increases with greater tension. For a standing wave of a fixed wavelength, the frequency of the fundamental resonance of the standing wave on a cord increases as wave speed increases.

Time spent:0 sec

Subject:
Physics

QID:404059

System:
Light and Sound